

ECLSS Closed-Loop Water Recycling & Condensate Distillation

1. Rotary Vacuum Distillation

Urine processor assemblies utilize vapor compression distillation within a spinning centrifugal drum to extract pure water vapor from wastewater brine at 93 percent efficiency.

2. Catalytic Post-Treatment

Condensate flows through high-temperature catalytic oxidizers operating at 120 degrees Celsius with zero-g compatible peristaltic pumps, eliminating volatile organic contaminants.

3. Mineralization & Biocide

Recycled water is re-mineralized with magnesium and calcium salts, dosed with ionic silver biocide at 400 ppb, and stored in dual 1500-liter titanium tanks.